

Math Thermo
class # 04
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The Euler Equation

Recall that internal energy and the entropy are homogeneous of degree one.

$$(4.1) \quad \tilde{U}(\lambda S, \lambda V, \lambda \vec{N}) = \lambda \tilde{U}(S, V, \vec{N}), \quad \lambda > 0,$$

where

$$\vec{N} = \begin{bmatrix} N_1 \\ N_2 \\ \vdots \\ N_r \end{bmatrix}.$$

Theorem (4.1): let $\tilde{U}$ be the internal energy of an isolated system. Then

$$(4.2) \quad \tilde{U} = TS - PV + \mu_1 N_1 + \dots + \mu_r N_r$$

Proof: Since $\tilde{U}$ is homogeneous of degree one differentiating equation (4.1) with respect to λ

we obtain

$$\begin{aligned} T(\lambda S, \lambda V, \lambda \vec{N}) S - P(\lambda S, \lambda V, \lambda \vec{N}) V \\ + \sum_{j=1}^r \mu_j(\lambda S, \lambda V, \lambda \vec{N}) N_j = \tilde{U}(S, V, \vec{N}). \end{aligned}$$

Taking $\lambda=1$, we get

$$\begin{aligned}\tilde{U}(S, V, \vec{N}) &= T(S, V, \vec{N})S - P(S, V, \vec{N})V \\ &\quad + \sum_{j=1}^r \mu_j(S, V, \vec{N}) N_j,\end{aligned}$$

as desired. ///

Remark: Equation (4.1) is known as Euler's Equation, and in some books it is called the integrated form of the internal energy.

Defn (4.2): A process path in state space Σ_S ,

$$\Sigma_S \subseteq [0, \infty) \times [0, \infty) \times [0, \infty) \times \cdots \times [0, \infty)$$

$\uparrow$
(S)

$\uparrow$
(V)

$\uparrow$
(N₁)

$\cdots$

$\uparrow$
(N_r)

is a continuous, piecewise smooth function $\vec{\gamma}: [0, 1] \rightarrow \Sigma_S$, defined by

$$\vec{\gamma}(\gamma) = \begin{bmatrix} S(\gamma) \\ V(\gamma) \\ N_1(\gamma) \\ \vdots \\ N_r(\gamma) \end{bmatrix}, \quad \gamma \in [0, 1].$$

A process path in Σ_V is defined similarly.

Thus, we have, using the chain rule,

$$(4.3) \quad \frac{d}{d\gamma} \tilde{U}(\vec{\gamma}(\gamma)) = T_0(\vec{\gamma}(\gamma)) S'(\gamma) - P_0(\vec{\gamma}(\gamma)) V'(\gamma) + \sum_{j=1}^r \mu_{0,j}(\vec{\gamma}(\gamma)) N_j'(\gamma)$$

for a valid process path in state space. Compare with

$$d\tilde{U} = TdS - PdV + \sum_{j=1}^r \mu_j.$$

Theorem (4.3): Suppose that $\vec{\gamma}: [0,1] \rightarrow \Sigma_S$ is a process path in state space Σ_S . Then,

$$(4.4) \quad 0 = S(\gamma) \frac{dT_0(\vec{\gamma}(\gamma))}{d\gamma} - V(\gamma) \frac{dP_0(\vec{\gamma}(\gamma))}{d\gamma} + \sum_{j=1}^r N_j(\gamma) \frac{d\mu_{0,j}(\vec{\gamma}(\gamma))}{d\gamma}.$$

This equation is called the Gibbs-Duhem relation.

Proof: Begin with the Euler equation and differentiate with respect to the process parameter γ :

$$(4.5) \quad \begin{aligned} \frac{d\tilde{U}}{d\gamma}(\vec{\gamma}(\gamma)) &= \frac{dT}{d\gamma}(\vec{\gamma}(\gamma)) S(\gamma) + T(\vec{\gamma}(\gamma)) S'(\gamma) \\ &\quad - \frac{dP}{d\gamma}(\vec{\gamma}(\gamma)) V(\gamma) - P(\vec{\gamma}(\gamma)) V'(\gamma) \\ &\quad + \sum_{j=1}^r \left\{ \frac{d\mu_j}{d\gamma}(\vec{\gamma}(\gamma)) N_j(\gamma) + \mu_j(\vec{\gamma}(\gamma)) N_j'(\gamma) \right\} \end{aligned}$$

Subtracting (4.5) from (4.3), we get (4.4). $\square$

Example (4.9): Suppose that the fundamental relation for a material is given by

$$\tilde{S} = 4A U^{1/4} V^{1/2} N^{1/4} + BN, \quad \Sigma_U = [0, \infty).$$

where $A, B > 0$ are constants. This function must be homogeneous of degree one. let us check that. Suppose $\lambda > 0$. Then

$$\begin{aligned} \tilde{S}(\lambda U, \lambda V, \lambda N) &= 4A \lambda U^{1/4} \lambda V^{1/2} \lambda N^{1/4} + B \lambda N \\ &= \lambda \tilde{S}(U, V, N) \quad // \end{aligned}$$

Recall that

$$\begin{aligned} T &= \frac{1}{\frac{\partial \tilde{S}}{\partial U}} = \left(A U^{-3/4} V^{1/2} N^{1/4} \right)^{-1} \\ &= \frac{U^{3/4}}{A V^{1/2} N^{1/4}}, \end{aligned}$$

which is homogeneous degree zero.

$$\frac{P}{T} = \frac{\partial \tilde{S}}{\partial V}$$

So

$$P = T \frac{\partial \tilde{S}}{\partial V}$$

$$= \frac{U^{3/4}}{A V^{1/2} N^{1/4}} \left(\frac{2A U^{1/4} N^{1/4}}{V^{1/2}} \right)$$

$$= \frac{2U}{V}$$

Finally,

$$\mu = -T \frac{\partial \tilde{S}}{\partial N}$$

$$= \frac{-U^{3/4}}{A V^{1/2} N^{1/4}} \left(\frac{A U^{1/4} V^{1/2}}{N^{3/4}} + B \right)$$

$$= -\frac{U}{N} - \frac{B U^{3/4}}{A V^{1/2} N^{1/4}} \quad ///$$

Example (4.5): Suppose that the fundamental relation is

$$\tilde{U} = \left(\frac{S - BN}{4 A V^{1/2} N^{1/4}} \right)^4$$

Recall that

$$T = \frac{\partial \tilde{U}}{\partial S}$$

$$= 4 \left(\frac{S - BN}{4 A V^{1/2} N^{1/4}} \right)^3 \frac{1}{4 A V^{1/2} N^{1/4}}$$

$$= \frac{\tilde{U}^{3/4}}{A V^{1/2} N^{1/4}},$$

the same as above.

$$P = -\frac{\partial \tilde{U}}{\partial V}$$

$$= -4 \left(\frac{S - BN}{4 A V^{1/2} N^{1/4}} \right)^3 \left(\frac{S - BN}{4 A N^{1/4}} \right) \left(-\frac{1}{2} \right) \frac{1}{V^{3/2}}$$

$$= 2 \left(\frac{S - BN}{4AV^{1/2}N^{1/4}} \right)^4 \frac{1}{V}$$

$$= \frac{2\tilde{U}}{V}.$$

Finally,

$$\mu = \frac{\partial \tilde{U}}{\partial N}$$

$$= 4 \left(\frac{S - BN}{4AV^{1/2}N^{1/4}} \right)^3 \frac{1}{4AV^{1/2}} \frac{N^{1/4}(-B) - (S - BN)\frac{1}{4}N^{-3/4}}{N^{1/2}}$$

$$= 4 \left(\frac{S - BN}{4AV^{1/2}N^{1/4}} \right)^3 \frac{1}{4AV^{1/2}} \frac{N^{1/4}(-B) - (S - BN)\frac{1}{4}N^{-3/4}}{N^{1/2}}$$

$$= 4\tilde{U}^{3/4} \cdot \frac{N^{1/4}}{4AV^{1/2}N^{1/2}} \cdot \left(-B - \frac{S - BN}{4N} \right)$$

$$= \frac{-B\tilde{U}^{3/4}}{AV^{1/2}N^{1/4}} - \frac{\tilde{U}^{3/4}}{AV^{1/2}N^{1/4}} \cdot \frac{(S - BN)}{4N}$$

$$= \frac{-B\tilde{U}^{3/4}}{AV^{1/2}N^{1/4}} - \frac{\tilde{U}}{N} \cdot \text{///}$$

let us establish the Euler Equation with respect to entropy.

Theorem (4.6): Let $\tilde{S}$ be the internal energy of an isolated system. Then

$$(4.6) \quad \tilde{S} = \frac{1}{T_s} U + \frac{P_s}{T_s} V - \sum_{j=1}^r \frac{\mu_{s,j}}{T_s} N_j,$$

where

$$T_s = T_s(U, V, \vec{N}),$$

$$P_s = P_s(U, V, \vec{N}),$$

$$\mu_{s,j} = \mu_{s,j}(U, V, \vec{N}),$$

and

$$\frac{1}{T_s} = \frac{\partial \tilde{S}}{\partial U}, \quad \frac{P_s}{T_s} = \frac{\partial \tilde{S}}{\partial V}, \quad \frac{\mu_{s,j}}{T_s} = \frac{\partial \tilde{S}}{\partial N_j}.$$

Proof: We again use the fact that $\tilde{S}$ is homogeneous of degree one. For any $\lambda > 0$,

$$\tilde{S}(\lambda U, \lambda V, \lambda \vec{N}) = \lambda \tilde{S}(U, V, \vec{N})$$

Taking the derivative with respect to λ , we have

$$\begin{aligned} & \frac{1}{T_s(\lambda U, \lambda V, \lambda \vec{N})} U + \frac{P_s(\lambda U, \lambda V, \lambda \vec{N})}{T_s(\lambda U, \lambda V, \lambda \vec{N})} V \\ & + \sum_{j=1}^r \frac{\mu_{s,j}(\lambda U, \lambda V, \lambda \vec{N})}{T_s(\lambda U, \lambda V, \lambda \vec{N})} N_j = \tilde{S}(U, V, \vec{N}) \end{aligned}$$

setting $\lambda = 1$ gives the desired result. //

The Gibbs - Duhem equation is similarly derived.

Theorem (4.7): Suppose that $\vec{\gamma}: [0,1] \rightarrow \Sigma_0$ is a process path. Then

$$(4.7) \quad 0 = \frac{dT_s}{d\tau}(\vec{\gamma}(\tau)) \tilde{S}(\vec{\gamma}(\tau)) - \frac{dP_s}{d\tau}(\vec{\gamma}(\tau)) V(\tau) + \sum_{j=1}^r \frac{d\mu_{sj}}{d\tau}(\vec{\gamma}(\tau)) N_j(\tau).$$

This equation is called the Gibbs - Duhem relation in the entropy form.

Proof: Using (4.6), we have

$$T_s(\vec{\gamma}(\tau)) \tilde{S}(\vec{\gamma}(\tau)) = U(\tau) + P_s(\vec{\gamma}(\tau)) V(\tau) - \sum_{j=1}^r \mu_{sj}(\vec{\gamma}(\tau)) N_j(\tau).$$

Taking the τ -derivative of the last equation we have

$$(4.8) \quad \begin{aligned} & \frac{dT_s}{d\tau}(\vec{\gamma}(\tau)) \tilde{S}(\vec{\gamma}(\tau)) + T_s(\vec{\gamma}(\tau)) \frac{d\tilde{S}}{d\tau}(\vec{\gamma}(\tau)) \\ &= U'(\tau) + \frac{dP_s}{d\tau}(\vec{\gamma}(\tau)) V(\tau) + P_s(\vec{\gamma}(\tau)) V'(\tau) \\ & \quad - \sum_{j=1}^r \left\{ \frac{d\mu_{sj}}{d\tau}(\vec{\gamma}(\tau)) N_j(\tau) + \mu_{sj}(\vec{\gamma}(\tau)) N_j'(\tau) \right\} \end{aligned}$$

Taking the τ derivative of $\tilde{S}(\vec{\gamma}(\tau))$ we have

$$\frac{d\tilde{S}}{d\tau}(\vec{\gamma}(\tau)) = \frac{1}{T_s(\vec{\gamma}(\tau))} U'(\tau) + \frac{P_s(\vec{\gamma}(\tau)) V'(\tau)}{T_s(\vec{\gamma}(\tau))}$$

$$- \sum_{j=1}^r \frac{\mu_{s,j}(\vec{s}(r))}{T_s(\vec{s}(r))} N_j'(r) .$$

Equivalently ,

$$(4.9) \quad T_s(\vec{s}(r)) \frac{dS}{dr}(\vec{s}(r)) = U'(r) + P_s(\vec{s}(r)) V'(r) - \sum_{j=1}^r \mu_{s,j}(\vec{s}(r)) N_j'(r)$$

Subtracting (4.9) from (4.8) we get

$$\frac{dT_s}{dr}(\vec{s}(r)) \tilde{S}(\vec{s}(r)) = \frac{dP_s}{dr}(\vec{s}(r)) V(r) - \sum_{j=1}^r \frac{d\mu_{s,j}}{dr}(\vec{s}(r)) N_j(r) ,$$

which is the desired result. ///

Remark: Compare (4.4) and (4.7):

$$(4.4) \quad 0 = S(r) \frac{dT_0}{dr}(\vec{s}(r)) - V(r) \frac{dP_0}{dr}(\vec{s}(r)) + \sum_{j=1}^r N_j(r) \frac{d\mu_{0,j}}{dr}(\vec{s}(r)) ,$$

$$(4.7) \quad 0 = \frac{dT_s}{dr}(\vec{s}(r)) \tilde{S}(\vec{s}(r)) - \frac{dP_s}{dr}(\vec{s}(r)) V(r) + \sum_{j=1}^r \frac{d\mu_{s,j}}{dr}(\vec{s}(r)) N_j(r) .$$

These are essentially the same expressions!

Suppose that $\tilde{S}$ is the fundamental entropy relation for an isolated system containing a unary ($r=1$) material.

$$\tilde{S} = \tilde{S}(U, V, N).$$

Suppose that $N > 0$ is fixed. Define

$$u := \frac{U}{N}, \quad \text{the molar energy,}$$

$$v := \frac{V}{N}, \quad \text{the molar volume.}$$

Then, since, for any $\lambda > 0$,

$$\lambda \tilde{S}(U, V, N) = \tilde{S}(\lambda U, \lambda V, \lambda N),$$

it follows that

$$\begin{aligned} \frac{1}{N} \tilde{S}(U, V, N) &= \tilde{S}\left(\frac{U}{N}, \frac{V}{N}, \frac{N}{N}\right) \\ &= \tilde{S}(u, v, 1). \end{aligned}$$

Now, define the molar entropy relation

$$\tilde{s} = \tilde{s}(u, v) := \tilde{S}(u, v, 1).$$

It then follows that

$$\tilde{S}(U, V, N) = N \tilde{s}(u, v).$$

likewise, if we define

$$s = \frac{S}{N} \quad v = \frac{V}{N}$$

and

$$\tilde{u} = \tilde{u}(s, v) := \tilde{U}(s, v, 1),$$

then it follows that

$$\tilde{U}(s, v, N) = N \tilde{u}(s, v).$$

Example (4.8) Suppose that

$$\tilde{S}(u, v, N) = \left(\frac{NVUR^2}{v_0 \theta} \right)^{1/3},$$

where $R, v_0, \theta > 0$ are constants. Then,

$$\begin{aligned} \tilde{S}(u, v) &= \tilde{S}(u, v, 1) \\ &= \left(\frac{vuR^2}{v_0 \theta} \right)^{1/3}. \end{aligned}$$

$$\begin{aligned} N\tilde{S}(u, v) &= \left(\frac{N^3 vuR^2}{v_0 \theta} \right)^{1/3} \\ &= \left(\frac{NVUR^2}{v_0 \theta} \right)^{1/3} \\ &= \tilde{S}(U, V, N). \end{aligned}$$

$\tilde{U}$ and $\tilde{S}$ are homogeneous of degree one (also called extensive variables).

T, P, μ on the other hand are homogeneous of degree zero (also called intensive variables).

Recall that, for all $\lambda > 0$,

$$T_0(S, V, N) = T_0(\lambda S, \lambda V, \lambda N).$$

Thus,

$$T_0(S, V, N) = T_0(s, v, 1)$$

We could, of course, give a new symbol for $T_0(s, v, 1)$, for example

$$t_0(s, v) := T_0(s, v, 1) \dots$$

Molar Euler and Gibbs-Duhem Equations

For a unary material ($r=1$), recall

$$\tilde{U}(S, V, N) = T_0(S, V, N)S - P_0(S, V, N)V + \mu_0(S, V, N)N$$

Suppose $N > 0$ is fixed. Then,

$$\frac{1}{N} \tilde{U}(S, V, N) = T_0(S, V, N) \frac{S}{N} - P_0(S, V, N) \frac{V}{N} + \mu_0(S, V, N)$$

or, equivalently,

(4.8)

$$\tilde{u}(s, v) = T_0(s, v) s - P_0(s, v) v + \mu_0(s, v).$$

This is the molar version of Euler's equation.

How do we get the Gibbs-Duhem equation?

Let $\vec{\gamma}(r) = (s(r), v(r))$ be a path in molar state space. Then

$$\frac{d}{dr} \tilde{u}(\vec{\gamma}(r)) = \frac{\partial \tilde{u}}{\partial s}(\vec{\gamma}(r)) s'(r) + \frac{\partial \tilde{u}}{\partial v}(\vec{\gamma}(r)) v'(r).$$

But

$$\frac{\partial \tilde{u}}{\partial s}(\vec{\gamma}(r)) \stackrel{\text{exercise}}{=} T_v(s(r), v(r))$$

and

$$\frac{\partial \tilde{u}}{\partial v}(\vec{\gamma}(r)) \stackrel{\text{exercise}}{=} -P_v(s(r), v(r)).$$

Thus, the combined First and Second laws are

(4.9)

$$\frac{d}{dr} \tilde{u}(\vec{\gamma}(r)) = T_v(\vec{\gamma}(r)) s'(r) - P_v(\vec{\gamma}(r)) v'(r).$$

On the other hand, differentiate the molar Euler equation (4.8), we have

(4.10)

$$\begin{aligned} \frac{d}{dr} \tilde{u}(\vec{\gamma}(r)) &= \frac{d}{dr} T_v(\vec{\gamma}(r)) s(r) + T_v(\vec{\gamma}(r)) s'(r) \\ &\quad - \frac{d}{dr} P_v(\vec{\gamma}(r)) v(r) - P_v(\vec{\gamma}(r)) v'(r) \\ &\quad + \frac{d}{dr} \mu_v(\vec{\gamma}(r)). \end{aligned}$$

Therefore, the molar Gibbs-Duhem equation is

(4.11)

$$0 = \frac{d}{dr} T_v(\vec{\gamma}(r)) s(r) - \frac{d}{dr} P_v(\vec{\gamma}(r)) v(r) + \frac{d}{dr} \mu_v(\vec{\gamma}(r))$$